ContextPrior-PLM for Stability Prediction

1. Input & Frozen PLM

Protein Sequence

M S E K L R Q A ⋯
L K K I E

Frozen PLM Encoder
(ESM-2 650M)

Residue-level Embeddings

1
2
⋮
L

$H\phi \in R^{\{L \times d\phi\}}$

Frozen (no update)
· Trainable

2. Local-Context Representation

Sliding windows

M_1
M_2
M_3
M_T

Window length F = 10
Stride s = 5
Max windows T_{max}=15

Local-context units

H_{e1}
H_{e2}
H_{e3}
H_{eT}

Local-context encoder ρ(·)

$H_e \in R^{\{T \times F \times d_m\}}$

Unit states

Z_1
Z_2
Z_3
Z_T

$Z \in R^{\{T \times d_m\}}$

3. Similarity Graph & Injection

Similarity graph

Edge weight
Low — High

default: NV descriptor
alternative: PLM similarity

Gated residual injection

Graph operator

$G(Z, A)$

$Z' = Z + G(Z, A)$

$Z' \in R^{\{T \times d_m\}}$

graph-enhanced states Z'

4. Prediction Head

Graph-based aggregation

Inter-unit message passing

Axial encoder (dual attention)

Along unit axis (T)
Along slot axis (F)
⊕

Captures interactions across units

Readout

Pooling
MLP

Predicted stability score
$\hat{y}$